Permutations and combinations

Explain how to count in each of the following situations. I have provided the number you should end up with. Your goal is to identify the counting strategy using tools such as the multiplication rule, permutations, and combinations.

How many possible batting orders are there for nine baseball players? (362,880)

16 horses race in the Kentucky Derby. How many possible results are there for win, place, and show (first, second, and third)? (3,360)

A tourist wants to visit six of America’s ten largest cities. In how many ways can she do this if the order of her visits is (a) important or (b) not important. (part (a) 151,200; part (b) 210)

In a class of 19 students, 7 will get A's. In how many ways could we pick these students? (50,388)

How many license plates are possible if the first three places are occupied by letters and the last three by numbers? (17,576,000) How many are possible if the three letters are all different and the three numbers are all different? (11,232,000)

The Duke basketball team has 10 women who can play guard and 12 women who can play the other three positions. At the start of the game, the coach gives the referee a starting lineup that lists who will play left guard, right guard, left forward, center, and right forward. In how many ways can this be done? (118,800)